Quantitative Decision Making

Graduate Course Project — Q2 2026

GlobalFlow Under Pressure

Network Design, Disruption, and Resilience in a Multi-Commodity Logistics Network

Group Project — Two Students per Group

Implementation: FICO Xpress Mosel or Python + Xpress

1. Context and Motivation

GlobalFlow is an international third-party logistics operator (3PL) that manages the end-to-end distribution of goods across a global network of suppliers, hubs, regional warehouses, and final customers. Unlike vertically integrated manufacturers, GlobalFlow does not own the goods it moves; it provides transportation, warehousing, and routing services to clients in multiple industrial sectors simultaneously.

This multi-sector exposure is precisely what makes GlobalFlow's situation strategically interesting. The company's network must serve clients with very different supply chain structures, subject to geopolitical and economic shocks that affect each sector in a different way and with different intensity. The current global environment makes this challenge particularly acute:

- The Russia–Ukraine conflict and ongoing tensions in the Iran region have caused sharp and sustained increases in energy costs, raising transportation expenses across all modes — with air freight most affected.
- The Taiwan Strait crisis has disrupted semiconductor supply chains, threatening the availability of key East Asian hub capacity.
- South Korea's export restrictions on certain technology components have constrained specific trade corridors.
- The reimposition of tariffs under US trade policy (informally referred to as "Trump-era tariffs") has raised costs significantly on transatlantic and transpacific flows.

GlobalFlow currently serves clients in three distinct product categories, each with its own demand structure, routing preferences, and exposure to external shocks:

- **Product A — Fertilizers and agricultural chemical precursors:** bulk or semi-bulk flows primarily transported by sea and road, with demand concentrated in European and emerging-market agricultural regions. Highly sensitive to energy cost shocks and port disruptions.
- **Product B — Semiconductors and electronic components:** high-value, low-weight goods transported predominantly by air. Heavily exposed to East Asian hub disruptions and technology export restrictions.
- **Product C — Battery components and energy transition materials:** medium-weight goods (lithium compounds, rare earth elements, battery cells) transported by a mix of sea and air depending on urgency. Affected by both tariff shocks and energy cost increases.

The combination of these three product categories on a shared network infrastructure creates rich trade-offs: a shock that disrupts Product B flows may free up capacity that can be redirected to Products A or C; a tariff increase on one corridor may make an alternative routing cost-effective for all three products simultaneously.

> **Central Question**
> *Is a network that is optimal under normal operating conditions also robust to the geopolitical and economic shocks that characterize today's global trade environment? And if not, at what cost can resilience be achieved?*

2. Network Structure

The GlobalFlow logistics network is represented as a directed graph G = (N, A), where N is the set of nodes and A is the set of directed arcs. The network has a four-tier hierarchical structure.

2.1 Node Types

- **Supplier Nodes ($S \subset N$), $|S| = 9$:** Origin nodes representing raw material production sites. Suppliers feed directly into the hub layer via always-active sea arcs (no fixed activation cost). Each supplier provides one specific product. Suppliers are grouped by product:

- **Product A (Fertilizers):** Minsk (potash/Belarus), Casablanca (phosphates/Morocco), Port of Spain (ammonia/Trinidad).
- **Product B (Semiconductors):** Taipei (logic chips/Taiwan), Seoul (memory/South Korea), Yokohama (sensors/Japan).
- **Product C (Battery components):** Antofagasta (lithium/Chile), Lubumbashi (cobalt/DR Congo), Chengdu (battery cells/China).

- **International Hubs ($H \subset N$), $|H| = 5$:** High-capacity transshipment nodes located at major intercontinental junctions, namely Frankfurt, Dubai, Singapore, Chicago, and São Paulo. Hubs do not generate or absorb demand; they serve exclusively as transit points. Hubs are always open but are subject to variable handling tariffs that change across scenarios.

- **Regional Warehouses ($W \subset N$), $|W| = 15$:** Intermediate storage and distribution nodes, each with a fixed opening cost and a capacity limit. These are located in: Rotterdam, Madrid, Warsaw, Mumbai, Chennai, Osaka, Tokyo, Shenzhen, Sydney, Houston, Toronto, Montreal, Monterrey, Manaus, and Bangalore. A warehouse may be open or closed; the decision to open or close a warehouse is a strategic binary decision. Only open warehouses can handle and forward flows.

- **Customer Nodes ($C \subset N$), $|C| = 30$:** Demand nodes representing final delivery destinations. Each customer has a fixed, known demand for each product. Customers can only receive goods through at least one open warehouse. Distribution: 8 customers in USA/Canada; 8 in Europe; 5 in Brazil; 5 in Japan; 4 in India.

The total network has $|N| = 59$ nodes. Sets S, H, W, and C are pairwise disjoint.

2.2 Arc Types and Cost Structure

Arcs A connect nodes across and within tiers. Not all inter-tier connections exist: the arc set reflects a realistic sparse connectivity structure (not a complete bipartite graph). Each arc $(i, j) \in A$ is characterized by:

- A variable transportation cost per unit of flow per product: c^p_{ij} (dependent on distance, transport mode, and product type).

- A shared capacity u_{ij} across all products (the arc can carry at most u_{ij} total units simultaneously, regardless of product mix).
- An optional fixed activation cost $f_{ij} > 0$ for arcs that require contracting a dedicated transport service or opening a new corridor. Supplier-to-hub arcs always have $f_{ij} = 0$ (always active). If $f_{ij} = 0$, the arc is available at variable cost only.

Transportation costs reflect both distance and modal choice: air freight is faster but significantly more expensive per unit; sea freight is cheapest for bulk flows; road freight is intermediate. The product type further modulates costs (e.g., weight/volume ratio, insurance, handling requirements).

2.3 Products

The three products P = {A, B, C} share the network infrastructure but have different cost and demand profiles. Each product $p \in P$ has:

- A set of dedicated supplier nodes $S^p \subset S$ that produce and inject supply into the network via hub connections.
- A demand d^p_j at each customer node $c \in C$ (units per planning period).
- Product-specific variable costs c^p_{ij} on each arc, reflecting mode affinity, weight/volume ratio, and regulatory requirements.

All data (node attributes, arc costs and capacities, product demands, supply quantities, tariff zones, and scenario parameters) are provided in the file globalflow_instance.xlsx. Students should load this file directly into their solver without modifying it.

3. Phase 1 — Optimal Network Design

3.1 Objective

Given a reference tariff scenario (the baseline scenario, representing current average operating conditions), determine the network configuration that minimizes total cost while satisfying all customer demands. This involves simultaneously deciding:

1. Which regional warehouses to open (strategic facility location decisions).
2. Which arcs with fixed activation costs to activate (network design decisions).
3. How to route each product's flows across the network (operational routing decisions).

3.2 Model

You are required to identify the nature of this network design problem, to formulate it, and to solve it using either FICO Xpress Mosel or Python+Xpress. The choice of formulation must be justified in your report. The formulation must include at minimum:

- Flow conservation constraints for each product at each node (sources, transshipment nodes, sinks).
- Shared capacity constraints on arcs linking arc flows across all products to the arc capacity and activation variable.
- Facility activation constraints ensuring that flows can only pass through open warehouses.
- Warehouse capacity constraints bounding total inbound flow at each open warehouse.
- Binary variables for warehouse opening and arc activation decisions.
- Non-negativity constraints on all flow variables.

In the case the model should prove too slow to solve at optimality, provide information on the maximum runtime allowed for the solver and justify it.

3.3 Formulation Analysis and Output of Phase 1

You are required to analyze the quality of your formulation by computing and reporting the LP relaxation bound and the LP-IP gap. If you have considered alternative formulations (e.g., aggregated vs. disaggregated capacity constraints, different linking constraints between flow and binary variables), compare their LP bounds and discuss which formulation produces the tightest relaxation and why. This analysis must appear in your report and will be evaluated as part of the MIP modeling component.

The output of Phase 1 is the baseline optimal solution, consisting of:

- The set of open warehouses $W^* \subseteq W$ and their utilization levels.
- The set of active arcs $A^* \subseteq A$.
- The optimal routing plan: flow x^p_{ij} for each product p on each active arc (i,j).
- The total optimal cost Z^* decomposed into fixed costs (warehouse opening + arc activation) and variable transportation costs.
- A visualization of the network and the output flows on the network.

4. Phase 2 — Stress Testing and Robustness Analysis

4.1 Objective

The baseline network $G^* = (N, A^*)$ designed in Phase 1 is now subjected to a structured battery of six adverse scenarios. For each scenario, you must evaluate how the optimal network responds and determines the best course of action.

4.2 Response Strategies

For each adverse scenario, you must evaluate three possible response strategies and determine which minimizes total cost under the new conditions:

- **Strategy R — Rerouting only:** Keep the baseline configuration (W^*, A^*) fixed. Only re-optimize the flow routing under the new cost/capacity conditions. Fixed costs have already been paid and cannot be recovered (they are "sunk").
- **Strategy A — Adaptation:** Allow partial reconfiguration of the network: open or close warehouses and activate or deactivate arcs, paying the associated fixed costs. This represents a medium-term operational adjustment.
- **Strategy F — Full redesign:** Re-solve Phase 1 from scratch using the scenario's cost and capacity parameters. This represents a long-term strategic response and gives a lower bound on achievable cost under the scenario, but ignores sunk and transition costs.

4.3 Adversarial Scenarios

The six scenarios are provided in globalflow_instance.xlsx (one sheet per scenario). Each scenario modifies the cost or capacity parameters of the baseline instance. The table below summarizes the six scenarios.

ID	Type	Description	Products Affected
T1	Tariff shock	Trump-era tariff increase (+40%) on all arcs crossing the transatlantic and transpacific trade corridors	B (semiconductors), C (battery components)
T2	Energy crisis	Asymmetric cost increase driven by Ukraine/Iran conflict: air freight +60%, road +20%, sea +10%	All products (intensity varies by route)

ID	Type	Description	Products Affected
T3	Seasonal surcharge	Airport and port handling fees rise +30% on all Hub nodes during peak season (Q4)	A (fertilizers via sea), B (air-heavy routes)
S1	Node disruption	Closure of the primary East Asian international hub (Singapore, simulating Taiwan Strait crisis)	B (semiconductors), C (battery components)
S2	Arc disruption	Blockade of the Korea-to-Europe export corridor (South Korea export restrictions)	B (semiconductors)
S3	Combined shock	Simultaneous application of T2 (energy crisis) and S1 (East Asian hub closure)	All products

S4 — Free scenario (group design): Design and implement one additional adversarial scenario of your choice. The scenario must be motivated by a real or plausible geopolitical or economic event, must modify at least one structural parameter of the network (costs, capacities, or node/arc availability), and must be guaranteed feasible. Describe the scenario in your report, justify its relevance, implement it in your solver, and analyze it using the same R/A/F framework as Scenarios T1–S3.

4.4 Required Analysis

For each of the six scenarios, your report must provide:

4. The optimal response strategy (R, A, or F) and the reasoning behind the choice.
5. The disruption cost: $\Delta Z = Z(\text{scenario, strategy}) - Z^*$.
6. A cost decomposition showing how the disruption cost breaks down across fixed and variable components.
7. An assessment of which products are most affected and why.

Across all six scenarios, your report must also address the following **strategic questions**:

- Which scenario is most damaging to GlobalFlow's network, and why?
- Is there a consistent structural weakness in the baseline network? (e.g., over-reliance on a single hub, a bottleneck corridor, a warehouse that becomes critical under multiple scenarios)
- What is the cost of resilience? Could a slightly more expensive baseline network (designed with robustness in mind) have significantly reduced disruption costs across scenarios?

5. Deliverables

5.1 Technical Report (PDF)

A report of no more than 10 pages, written in latex, in English, and in two-column format. The report must include an abstract and must cover:

8. The complete MIP formulation used in Phase 1, with clear definition of all variables, parameters, and constraints.
9. Phase 1 results: the optimal configuration, cost decomposition, and a commentary on the structure of the solution.
10. Phase 2 results: for each scenario, the optimal strategy, disruption cost, and analysis. A summary table is expected.
11. Strategic conclusions: answers to the three strategic questions in Section 4.4, supported by quantitative evidence from your results.

5.2 Code

All source code must be submitted as a separate attachment (not counted in the 10-page limit). The code must be organized into clearly named modules:

- phase1_solver.py / phase1.mos: the Phase 1 MIP model and solver.
- phase2_solver.py / phase2.mos: the Phase 2 multi-strategy solver.
- analysis.py (optional): scripts for result visualization and table generation.

Code must be commented sufficiently for a reader to understand the structure and the correspondence between code and mathematical formulation. All results in the report must be reproducible by running the submitted code.

6. Evaluation Criteria

Component	Weight	What is Evaluated
MIP Model (Phase 1)	40%	Correctness of formulation, variable and constraint design, LP relaxation quality
Robustness Analysis (Phase 2)	40%	Depth of scenario comparison, quality of strategic reasoning, cost decomposition
Code Quality	20%	Modularity, readability, documentation, reproducibility

What distinguishes an excellent project

A mechanical project solves the MIP and reports numbers. An excellent project goes further: it asks why the solution looks the way it does, identifies the structural reasons behind the network's vulnerabilities, and uses the quantitative results to build a coherent strategic argument. The robustness analysis in Phase 2 is not a series of independent solver runs — it is a structured investigation of the tension between optimality and resilience.

Groups that demonstrate this kind of analytical depth will be rewarded accordingly, independently of whether their formulation is the most sophisticated possible.

7. Suggested Timeline

- Week 1: Study the problem and analyze the instance provided in globalflow_instance.xlsx.
- Week 2: Formulate and implement the Phase 1 MIP; solve and analyze the baseline solution.
- Weeks 3–4: Implement Phase 2; run all six scenarios; collect and interpret results.
- Week 5: Write the report and prepare the presentation.

Start with a small sub-instance (e.g., 2 hubs, 4 warehouses, 8 customers) to validate your formulation before scaling to the full network. Solver performance on MIP problems is highly sensitive to formulation quality — a weak formulation may be correct but take far too long to solve on the full instance.